

FasalKawach

The trust layer for India's ₹1.83 lakh crore crop insurance system — turning a one-year claim wait into a same-day, satellite-verified payout.

●The wedge

Localized, individual-farm damage claims — the one PMFBY category YES-TECH's area-level satellite model still can't automate.

●The engine

Farmer photo + satellite NDVI + weather data, fused into one insurer-ready confidence score.

●The revenue

Per-claim fee to insurers, who now face a statutory 12% annual penalty for slow settlement.

"22.67 crore Indian farmers have received a PMFBY payout since 2016 — and millions more are still waiting, with no way to prove their field was ever damaged. We give them that proof, instantly."

Theme	Stage	Prepared for	Season
AI/GenAI × Agritech × FinTech	Working MVP, 8-week build	Grand Finale Jury, LPU	BIC Season 2, Nov 2026

A satellite-verified evidence layer for crop insurance claims

FasalKawach turns a farmer's phone photo into an insurer-ready proof-of-damage packet in minutes, not months.

The unfair advantage. The moment a hailstorm, flash flood, fire, or wild-animal attack hits a farmer's field, FasalKawach fuses a geo-tagged farmer photo with free satellite vegetation data and government weather records into one evidence packet with a confidence and fraud-risk score — collapsing what today takes insurers weeks of manual field surveys into a same-day digital decision. We are the trust and speed layer sitting on top of India's ₹1.83 lakh crore crop insurance system.

Built inside the rules, not around them. Crop-loss assessment in India is a licensed function — IRDAI runs a dedicated "Crop Insurance" department under its Surveyors and Loss Assessors Regulations. So FasalKawach does not approve claims. It generates the evidence and the recommendation a licensed surveyor signs off on, cutting their turnaround from weeks to hours instead of trying to replace them. That distinction is the difference between a product an insurer can actually deploy and one their compliance team blocks on day one.

₹1.83L Cr Paid out under PMFBY since 2016, across 22.67 crore farmer claims	12%/yr Penalty insurers now pay for delayed claim settlement (from Kharif 2024)	\$974M India agritech market in 2025, headed to \$2.52B by 2034	\$145M+ Already invested in India's satellite-for-agriculture sector
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Why now. The Government of India has already spent three years and significant political capital proving that satellite-based yield assessment works — YES-TECH now carries mandatory weightage in claim calculations for paddy and wheat. But that system operates at the block/area level. The moment the damage is *localized* — a hailstorm on one farmer's two-hectare plot, an animal attack, a farm fire — the rulebook still requires an individual, on-the-ground survey. That is the exact segment nobody has automated. That is FasalKawach.

One-line pitch for the jury: "YES-TECH proved India that satellites can replace surveyors for area-level claims. We're the team that finishes the job for the 1-in-4 claims that are individual, localized, and still stuck on paper."

Business model in one sentence. We prove the model with one insurer, one district, one cropping season — then charge a per-claim fee for every evidence packet we generate, priced against a cost insurers are now legally required to avoid.

What we can demo in 8 weeks. A working pipeline that pulls real Sentinel-2 satellite imagery for a real Indian district, computes before/after vegetation stress, cross-checks it against public weather data, scores a sample farmer-submitted photo with a lightweight damage classifier, and outputs a fused confidence score on a live insurer-style dashboard — end to end, on stage, in under three minutes.

India's crop insurance has a trust crisis, not just a tech gap

Pradhan Mantri Fasal Bima Yojana (PMFBY) is the world's largest crop insurance programme by farmer count. It has paid out real money to real families. But the lived experience for a large share of farmers is still defined by delay, opacity, and disputes:

A soybean and wheat farmer in Sehore district, Madhya Pradesh, lost more than a third of his crop to unseasonal rain and animal attacks in the 2025 kharif season. A year later, he had received no update and no payment — despite his district being one of the states furthest along in adopting satellite-based yield assessment.

This is not an isolated anecdote. Official dashboards show total claim disbursement across three central Madhya Pradesh districts — Bhopal, Sehore, and Vidisha — falling from roughly ₹218 crore (kharif season, 2018) to about ₹55 crore (2024), even as the state became a flagship rollout site for remote-sensing-based assessment. Technology adoption and farmer trust have been moving in opposite directions.

Why the government's own fix doesn't close this gap. Since Kharif 2023, the Ministry of Agriculture's Yield Estimation System based on Technology (YES-TECH) has mandatorily assigned 30% weightage to satellite-derived yield data for paddy and wheat — a genuine, well-funded, credible reform. But by design and by policy text, PMFBY's area-approach and YES-TECH cover *widespread* seasonal risk. Damage from hailstorm, landslide, inundation, cloudburst, fire, and localized wild-animal or pest attack is explicitly assessed **on an individual insured-farm basis** — which today still means a human surveyor has to physically visit, weeks after the event, often after evidence has already washed away, burned, or been re-ploughed.

Claim type	Who assesses it today	Status
Area-wide seasonal yield shortfall	YES-TECH satellite + limited manual Crop Cutting Experiments	Automating, government-led, well-funded
Localized hailstorm / flood / fire / animal attack (individual farm)	Manual on-farm survey, mandated by policy	Untouched — this is our wedge

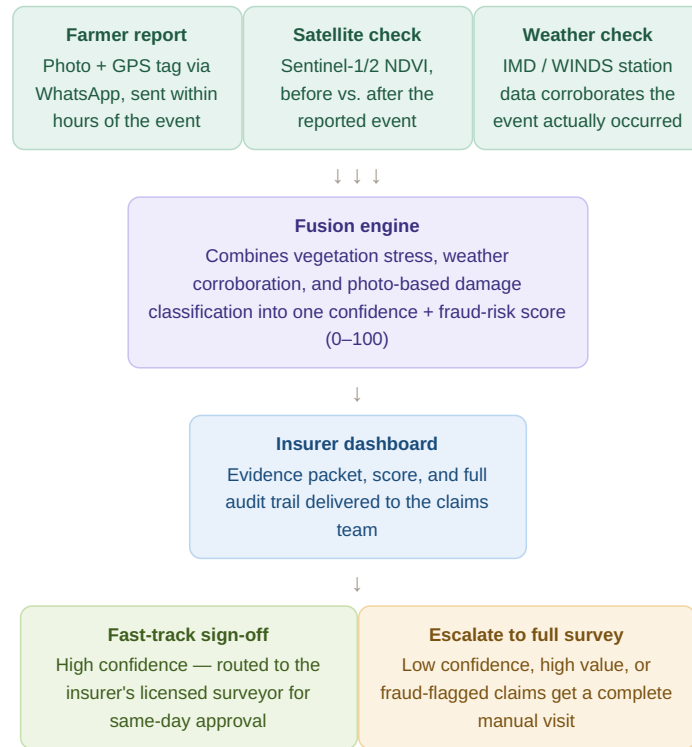
The Government of India is not blind to the trust problem — insurers now face a statutory **12% annual penalty** for delayed claim settlement, effective from Kharif 2024, and pilots like a WhatsApp-based PMFBY helpdesk in Chhattisgarh show the state actively experimenting with low-friction digital channels for exactly the population we're building for. The political will and the regulatory pressure both already exist. What's missing is the product that turns "a farmer's field just got destroyed" into a defensible, fast, digital proof — for the one category of claim the system was never built to automate.

The size of the wedge: official data shows over 78 crore farmer applications insured and 22.67 crore claims paid since PMFBY's launch. Even a conservative estimate that localized, individual-basis claims make up a meaningful minority of that volume puts the addressable claim-processing market in the tens of millions of events over the scheme's lifetime — recurring every single cropping season.

From farmer photo to insurer-ready evidence packet

A fusion engine that turns three weak, independent signals into one strong, defensible claim score

FasalKawach sits between the farmer and the insurer as an automated, always-on field surveyor. It does not replace YES-TECH's area-level yield model — it plugs the exact hole YES-TECH leaves open: individual, localized, event-triggered damage.



Why this is defensible as a claim, not just a demo. No single signal is trustworthy alone — a farmer photo can be staged, satellite NDVI can be confused by ordinary harvest timing, and weather station data is often too sparse to pinpoint a single 2-hectare field. Fusing all three is what makes the output something a licensed surveyor can act on with confidence, and it's the reason this can't be replicated by simply "adding AI" to one data source. The decision authority never leaves the surveyor — we make their decision faster and better evidenced, which is what actually gets this approved for real-world use instead of stalling in an insurer's compliance review.

STEP 1

Capture

Farmer sends a photo, short video, and one-tap event type (hailstorm / flood / fire / animal attack) over WhatsApp Business API. Location is captured via native share, no manual GPS entry.

STEP 2

Cross-verify

Backend pulls the registered plot polygon, fetches Sentinel imagery for before/after the reported date via Google Earth Engine, and checks public weather records for the same date and location.

STEP 3

Classify

A lightweight vision model trained on labeled crop-damage imagery classifies the farmer's photo by damage type and severity, independent of the satellite signal.

STEP 4

Fuse and route

A scoring model combines all three signals into one confidence and fraud-risk score. High-confidence claims are routed to the insurer's licensed surveyor for same-day sign-off; ambiguous or high-value ones go to a full manual visit, with the evidence packet already prepared.

What's actually running under the hood

Every component below is buildable and demoable inside an 8-week window

Layer	What it does	Tools
Farmer interface	Zero-download reporting channel, works on basic smartphones	WhatsApp Business Cloud API, conversational flow
Satellite intelligence	Pulls before/after imagery for the farmer's exact plot, computes vegetation-stress index	Google Earth Engine (free tier), Sentinel-2 optical NDVI for Phase 1; Sentinel-1 radar layered in for cloud-covered monsoon events in Phase 2
Weather corroboration	Confirms the claimed event actually occurred at that place and time	IMD/WINDS public data, open weather station APIs
Damage classifier	Scores the farmer's photo independently of the satellite signal	Transfer-learned lightweight CNN (MobileNet/EfficientNet base), trained on public crop-damage image sets plus self-collected samples
Fusion & scoring	Combines the three signals into one 0–100 confidence and severity score	Gradient-boosted tree / weighted scoring model, explainable by design
Insurer dashboard	Displays incoming evidence packets, sorted by confidence, with one-click approve or escalate	Web dashboard, PDF evidence-packet generator, simple claims-team login
Data layer	Stores every submission, satellite match, and eventual outcome as a labeled training example	Structured database that compounds into a proprietary farm-level damage dataset over time

Demo-day build plan: pick one real Indian district and crop season with a documented, clear-sky-verifiable event — hailstorm, fire, or animal attack, where optical NDVI is reliable — pull the actual satellite imagery for that date range, run the NDVI comparison live, and pair it with a small set of real or realistically staged farmer photos. The jury sees real data, not a mockup.

Scoped honestly. Phase 1 targets the claim types where optical satellite data is reliable: hailstorm, fire, and localized animal or pest damage. Cloudburst and flash-flood claims — which occur under exactly the cloud cover that blinds optical imagery — are a deliberate Phase 2, once Sentinel-1 radar processing is production-ready. We'd rather own a narrower claim well than a broad one badly.

What makes this technically credible, not just impressive-sounding: Sentinel satellite data is free and public, Google Earth Engine's Python/JavaScript API is extensively documented with existing NDVI crop-monitoring tutorials, and India's own Ministry of Agriculture has already validated that this class of remote-sensing approach works well enough to mandate it for area-level claims. We are applying a proven government-validated methodology to a smaller, harder, individually-verified case — that is a feature, not a risk, when pitching to a jury that includes insurers and agritech operators.

We engineered around the hard parts before anyone had to ask

The four questions a sharp insurance-industry judge will ask, answered before they're asked

REGULATORY

"Who's actually authorized to approve this claim?"

Not us. Crop-loss assessment is a licensed IRDAI function. FasalKawach is designed as a co-pilot for licensed surveyors, not a replacement — we generate the evidence packet and confidence score; a licensed surveyor retains sign-off authority on every claim, fast-tracked or not. This is also why insurers can actually deploy us: we reduce their compliance risk instead of creating new liability for them.

INCENTIVE ALIGNMENT

"Why would an insurer pay to make it easier for farmers to win against them?"

Because the fraud-risk score cuts both ways. The same fusion engine that fast-tracks a genuine hailstorm claim also flags a staged photo or a mismatched plot — protecting insurer margins, not just farmer outcomes. We're not pitched as a farmer-advocacy tool; we're pitched as a claims-integrity tool that happens to also make genuine claims faster. That's what gets budget approved.

DATA QUALITY

"What about India's messy land records and tiny, fragmented plots?"

We don't pretend satellite resolution solves everything alone. For plots too small or fragmented for a confident NDVI read, the fusion score weights farmer photo and weather corroboration more heavily and is designed to say "insufficient satellite confidence — escalate to manual survey" rather than force a bad automated call. An honest low-confidence flag is more valuable to an insurer than a falsely confident one.

DATA PROTECTION

"How are you handling farmer location, financial, and personal data?"

Farmer data (location, photos, bank details for payout) is personal data under India's Digital Personal Data Protection Act, 2023. Our architecture separates identity data from claim-evidence data, keeps raw farmer submissions out of any third-party sharing, and is designed for a formal data-processing agreement with each insurer partner before any pilot goes live — not an afterthought bolted on post-launch.

A pilot-first path to monetizing a statutory pain point

The single most important fact in this business model: as of Kharif 2024, insurers are **legally penalised 12% annually** for delayed PMFBY claim settlement. We are not asking insurers to buy something because it's nice to have — we are letting them avoid a cost the government now imposes on them by law, while also cutting their fraud exposure. That turns the sales conversation from "please try our product" into "here is what slow, unverified claims are costing you, and here is what we charge to fix it." We earn that conversation with a paid pilot before we earn a platform contract — not the other way around.

Customer	Model	Why they pay
General insurers (PMFBY-empanelled: AIC, ICICI Lombard, HDFC ERGO, Reliance General, and others)	Per-claim fee for each evidence packet generated; volume-tiered SaaS pricing	Avoids the 12% annual late-payment penalty; cuts manual surveyor cost and cycle time from weeks to hours
State agriculture departments	Platform licensing as a citizen grievance and verification layer feeding the National Crop Insurance Portal	Reduces farmer complaints, strengthens the political case for the scheme's credibility
Reinsurers & portfolio underwriters	Access to the aggregated, anonymized risk dataset for pricing and reserving	Better loss forecasting from granular, farm-level ground truth
Farmers	Free, forever	The free side of the network is what generates the proprietary dataset that makes the paid side valuable

The flywheel. Every claim processed adds a labeled data point — farmer photo, satellite signature, weather record, and eventual surveyor decision — that no one else has at this granularity. More farmers using the free channel produces more training data, which improves the fusion model's accuracy, which increases surveyor trust in the fast-track recommendation, which brings insurers back with more claim volume, which reaches more farmers through insurer-driven onboarding. That is a genuine data-network effect, not just a marketing phrase.

Path to first revenue: a single pilot agreement with one PMFBY-empanelled insurer in one district, running one cropping season, priced per evidence packet generated. That pilot is the proof point every subsequent state and insurer conversation gets built on.

Expansion surface. The same fusion-scoring engine generalises far beyond PMFBY: livestock-loss verification, parametric drought and heat micro-insurance for horticulture and floriculture crops that PMFBY under-covers, disaster-relief fund verification for state and central agencies, and — longer term — licensing the same playbook to other monsoon-dependent economies running comparable area-based crop insurance schemes.

A proven category, with one clear gap left unclaimed

This is not an unproven category we're hoping investors will believe in — Indian and global capital has already validated satellite-driven agri-intelligence as a fundable, scalable business:

Company / initiative	What they do	Where FasalKawach differs
YES-TECH (Government of India)	Satellite-based yield estimation, mandatory 30% weightage for paddy & wheat under PMFBY	Operates at block/area level; explicitly excludes individual localized-damage claims
CropIn (Bengaluru; \$43.8M raised across 5 rounds)	Enterprise farm-data intelligence platform across 30M+ acres, serving agribusinesses, banks, insurers, governments	Built for large agribusiness and portfolio-level analytics, not farmer-initiated, event-triggered individual claims
Pixxel (~\$96M raised)	Hyperspectral satellite imaging infrastructure	Infrastructure and imagery layer; not a claims-workflow or insurer-facing product
India space-tech-for-agriculture sector	~191 space-tech startups nationally, ~21 focused on satellite imaging/crop analytics/weather, ~\$145M+ disclosed sector funding	Validates capital appetite for exactly this data layer

\$974M → \$2.52B India agritech market, 2025 to 2034 (10.6% CAGR)	78.4 Cr Farmer applications insured under PMFBY since 2016	22.67 Cr Claims already paid out under the scheme
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The whitespace, stated precisely: every comparable player operates at the aggregate, area, or infrastructure level. None of them are building the farmer-initiated, event-triggered, individually-verified evidence layer for the specific claim category that PMFBY's own rulebook still requires a human to assess. That is a narrow wedge by design — narrow enough for a small team to own convincingly, and connected to a market large enough to expand from once owned.

What we know, what we don't yet, and how we close the gap

We'd rather show you an honest plan than a fabricated Letter of Intent

We have not yet signed an insurer pilot — we're a student team, eight weeks into a build, not a company with a sales cycle behind it. Any pitch that implies otherwise wouldn't survive one follow-up question, and claiming false traction would violate this competition's own originality rules. Here is exactly where we stand and exactly what closing the gap looks like:

Stage	What it involves	Timeline
Technical proof	Working pipeline on real Sentinel data for one real district and one documented event, shown live at the Grand Finale	Complete by finale
First conversations	Structured outreach to district-level insurance surveyors, agri-insurance TPAs, and PMFBY-empanelled insurer regional offices — to pressure-test the fusion score against a real surveyor's judgment, not to sell yet	Weeks 1–3 post-finale
No-cost pilot proposal	A one-district, one-season pilot offered free to a willing insurer or state agriculture department, in exchange for real claim data to validate the fusion score against actual outcomes	Next cropping season
Paid pilot	Convert validated pilot results into the first per-claim commercial agreement	Post-validation

Why this sequencing is the credible answer, not a weaker one: a jury of real VCs and insurers has seen hundreds of teams claim traction they don't have. A team that says "here's our evidence pipeline working on real data, and here's exactly how we'll validate it with a real surveyor before we ask anyone for money" reads as more fundable, not less — because it shows we understand how insurance actually works, not just how AI demos work.

The jury hook

1 We designed for the regulation instead of hoping nobody asks.

Most student teams pitching "AI for insurance" haven't checked whether claims assessment is a licensed activity in India. We have, we know it is, and we built FasalKawach as a surveyor's co-pilot from day one — not because we got caught out and patched it, but because that's the only version of this product an insurer's legal team lets through the door.

2 The moat is the dataset, not the model.

Every submission — farmer photo, satellite match, weather corroboration, and eventual surveyor decision — becomes a labeled ground-truth record at a granularity (individual farm, individual event) that no government system or existing agritech company currently collects. That dataset compounds with every claim season and becomes licensable in its own right, the same way proprietary risk data is monetized by global reinsurance-data players.

3 The regulatory tailwind is already in motion.

We don't have to convince anyone that claim speed matters — the Government of India already legislated a 12% annual penalty for slow insurers, starting Kharif 2024. The ROI conversation with a paying customer is arithmetic, already written into policy, before we say a single word.

4 We complete a story India has already invested in telling.

YES-TECH is a genuine, well-funded government success story for area-level claims. We are not competing with it — we are the team that finishes the story for the individual, localized claims it was never designed to solve. That framing turns skepticism about "another agritech idea" into recognition of a specific, well-evidenced gap.

5 One playbook, many markets.

The same fusion engine extends to livestock insurance, disaster-relief verification, parametric micro-insurance for crops PMFBY under-covers, and export to other monsoon-economies running comparable area-based schemes — a single technical core with several independent, multi-billion-dollar expansion paths.

6 The on-stage demo is genuinely visual.

A live before/after satellite heatmap flip, a real WhatsApp submission triggering a real-time score, and a payout notification — a jury watches the entire value proposition happen in under two minutes, using real data, not slides.

A note on our numbers: "farmer applications insured" (policies issued) and "claims paid" (successful payouts) are different metrics from different points in PMFBY's lifecycle — we've kept them labeled separately throughout this document rather than blending them into one bigger-sounding figure.

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